Metric Trees, PM Vocabulary & Case Scenarios

Analytics / Product interview study guide
Hike · eBay · Expedia · Flipkart · Groww

> *Every case answer follows one pattern:* ***clarify → decompose the metric tree → state 3–4 hypotheses → name the single data cut that confirms or kills each → prioritise by size × ease → recommend and name the success metric.***

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1. Metric tree examples

Multiplicative trees (factors multiply) let you attribute a % change in the top metric to each factor: Δ%GMV ≈ Δ%Base + Δ%Conv + Δ%TPC + Δ%AOV (exact in logs). **Additive trees** (parts sum) suit segment splits and flows such as New + Resurrected − Churned.

Business	Tree
E-commerce GMV	Active base × Conversion (% transacting) × TPC (txns per buyer) × AOV
Same, deeper	AOV = Items per order × ASP × (1 − discount%). Base = New + Retained + Resurrected
Marketplace revenue (eBay)	GMV × Take rate. GMV = Active buyers × Frequency × ASP. Supply view: Sellers × Listings/seller × Sell-through × ASP
Broker (Groww, F&O/stocks)	Active traders × Orders/trader × Avg order value × Brokerage/order
Broker (MF)	Revenue = AUM × Take rate. AUM = Opening + Net flows (SIPs × SIP size − Redemptions) + Market movement
Broker acquisition funnel	Installs × Signup% × KYC complete% × First funding% × First trade%
Ads	Impressions × CTR × CPC. Impressions = Sessions × Ad slots × Fill rate
Hike Cinema watch time	DAU × Sessions/DAU × Episodes/session × Min/episode (Episodes/session = Autoplay continuation × Completion)
Subscription	Revenue = Subs × ARPU. Subs(t) = Subs(t−1) + New + Reactivated − Churned
Promo ROI	(Incremental GMV × Take rate) − Promo cost. Incremental GMV = Redeemers × Uplift per redeemer
Loyalty (Plus / Netflix-style)	Plus GMV = Members × % active × Orders/active member × AOV. Members = Enrolled − Lapsed
Retention (messaging)	MAU = Retained + New + Resurrected. DAU/MAU = Active days per user ÷ 30
Quick commerce / food (new)	Orders = Active users × Frequency. Frequency driven by Availability (fill rate) × ETA × Delivery-fee threshold. Revenue = Orders × (AOV × commission + delivery fee + ad income) − Cost per order
Payments / UPI (new)	TPV = Active users × Txns/user × Avg ticket. Successful txns = Attempts × (1 − Failure rate). Share of wallet = Own TPV ÷ User total digital spend
Travel / Expedia (new)	Bookings = Searches × Search-to-book%. Revenue = Bookings × (1 − Cancel%) × Booking value × Take rate + Attach (hotel+flight) revenue

Why we build metric trees

- **Localise a movement:** turns "GMV dropped" into a specific, testable branch instead of a random search.
- **Ownership:** each leaf maps to a team (growth owns base, product owns conversion, pricing owns AOV), so no orphan metrics.
- **Input vs output metrics:** leadership watches the root, teams pull the leaves, OKRs are set on levers rather than outcomes.
- **Sizing and forecasting:** "+2% conversion is worth +X% GMV"; set targets bottom-up.
- **Avoids misleading aggregates:** exposes mix effects, e.g. AOV up only because low-end buyers left (Simpson's paradox).
- **Shared language:** a written tree is the metrics glossary; stops definition arguments.

> ***Pitfalls to mention:*** *factors are rarely independent (a promo raises conversion and lowers AOV), so attribute carefully. Don't over-decompose beyond the level where someone can act. Always include a guardrail branch (e.g. opt-outs, support contacts, cancellations).*

2. Decomposition questions with answers

Q. GMV is down 8% WoW. Walk me through it.

Split into Base × Conv × TPC × AOV and find which factor moved, then cut that factor by new vs existing, platform and category. Base ↓ = acquisition or retention; Conv ↓ = funnel or pricing; TPC ↓ = habit or promos; AOV ↓ = mix or discounting. First rule out tracking error and seasonality (YoY).

Q. Orders are up 12% but GMV is flat.

AOV or mix shift: more low-ASP categories, deeper discounts, or a promo pulling small baskets. Check category mix to rule out Simpson's paradox.

Q. Groww brokerage revenue fell while new accounts grew.

Active traders × Orders/trader × AOV × Brokerage/order. Likely new users are inactive or low-frequency: check activation (first trade) and F&O vs cash mix. A SEBI rule change on F&O would hit orders/trader.

Q. Cinema watch time +20%, D30 retention flat.

Break into DAU × sessions × episodes × minutes. If the lift is minutes/episode or autoplay among existing users, it is depth among the already-retained, not a new habit. Track retained-viewer count, not total minutes.

Q. Marketplace revenue is flat. Buyers or sellers?

Revenue = GMV × Take rate. Check whether take rate changed (mix, fee discounts, ads share). Then split GMV into demand (buyers × freq × ASP) and supply (listings × sell-through).

Q. Build a metric tree for Hike Messenger's north star, Daily Sending Users.

DSU = DAU × % who send. DAU = New + Retained + Resurrected. Retained is driven by contact density and groups (input metrics). Name an owner per branch.

Q. Quick commerce: orders per user fell but AOV rose.

Frequency × AOV. Likely low-frequency users left (survivorship), or a delivery-fee threshold pushed bigger, rarer baskets. Check cohort frequency and basket distribution around the threshold; availability/ETA by dark store.

Q. UPI app: TPV flat but active users up.

TPV = Users × Txns/user × Ticket. New users are low-activity or ticket size fell. Check first-transaction rate, failure rate by bank/handle, P2P vs merchant mix.

3. PM vocabulary by industry

Universal funnel words: awareness → **intent** → **visibility / discovery** → **consideration** → conversion → **activation** (aha moment) → habit → retention → resurrection → advocacy. Also: friction, drop-off, cold start, stickiness, cohort, power users, leading vs lagging indicator, North Star, guardrail, input vs output metric, incrementality, counterfactual, mix shift.

Industry / app	Words to drop	Example sentence
E-commerce (Flipkart, Amazon)	Search intent, discovery, PDP view, consideration (wishlist, compare), add-to-cart, checkout friction, price perception, deal density, assortment, delivery promise, basket building, repeat rate	"Visibility is fine but consideration is weak: PDP-to-cart dropped, so I'd check price perception and delivery promise."
Marketplace (eBay)	Liquidity, supply/demand balance, sell-through, take rate, cold start, trust & safety, ranking/visibility, leakage, enthusiast buyers, additionality	"Supply is healthy but sell-through is low, so it's a demand-side liquidity problem."
Broker / wealth (Groww, Zerodha)	KYC completion, first funding, first trade (activation), active trader, order frequency, AUM, SIP stickiness, cross-sell ladder (MF → stocks → F&O), risk profiling, drawdown churn, regulatory headwind	"Funnel leaks between KYC and first funding, so it's intent decay, not acquisition."
Messaging (Hike, WhatsApp)	Connected user, social graph, magic number, sender vs reader, network effect, invite loop, contact density	"Retention here is social: I'd find the magic number of active contacts."
Micro-drama / OTT (Hike Cinema, Netflix)	Hook, cliffhanger, time-to-first-play, completion, autoplay continuation, binge, series completion, content-market fit, watchlist, win-back	"Ep1 completion is the hook, ep2 within 48h is the habit loop."
Loyalty / subscription (Plus, Expedia, Prime)	Enrollment, activation, tenure, streak, earn/burn, tier progression, breakage, benefit utilisation, perceived value, lapse, involuntary vs voluntary churn	"Enrollment isn't value: I'd track benefit utilisation and burn rate."
Travel (Expedia, MMT)	Search-to-book, shop vs book, abandonment, price lock, attach rate, cancellation, repeat booking, seasonality, lead time	"Look-to-book fell, so I'd split by lead time and route before blaming UX."
Quick commerce / food (Blinkit, Swiggy)	Serviceability, fill rate, ETA, dark-store density, order frequency, basket size, delivery-fee threshold, cancellation/RTO	"Frequency is driven by availability and ETA, so check fill rate first."
Payments (UPI, PhonePe)	Bank linking, first transaction, P2P vs merchant, share of wallet, failure rate, TPV, cashback dependence	"Activation is linking plus first successful transaction; failure rate is the guardrail."
Ads	Impression, CTR, CPC, fill rate, ad load, eCPM, incrementality, attribution, lift	"Last-touch overstates, so I'd use a holdout for incrementality."
Ride-hailing (Uber, Ola) (new)	Supply-demand balance, driver utilisation, ETA/ETD, surge, cancellation (rider vs driver), wait time, match rate, incentives, repeat rider	"Cancellations are a supply-side signal: I'd check driver ETA and acceptance rate by zone."
Digital lending (new)	Application-to-disbursal, credit policy, approval rate, NPA/delinquency, roll rate, cost of acquisition, cross-sell, repeat borrower, bureau pull	"Approval rate up with roll rate up means we loosened risk, not grew good volume."
Social / content feed (Instagram, ShareChat) (new)	Creator-consumer loop, feed relevance, time spent, session depth, follow graph, content supply, virality, notification CTR, quality vs quantity	"Time spent rose but sessions fell: that is feed depth, not habit."

Industry / app	Words to drop	Example sentence
EdTech (new)	Trial-to-paid, lesson completion, streaks, cohort completion, mentor load, refund rate, learning outcome, drop-off by module	"Completion drops at module 3, so it is a content-difficulty issue, not acquisition."
Fantasy / real-money gaming (new)	Deposit conversion, first-contest, contests per user, bonus dependency, responsible-gaming guardrails, ARPPU, cohort LTV, regulatory risk	"Engagement is bonus-led; I'd judge it on non-bonus deposits per cohort."

4. Fifteen PM case scenarios: hypotheses and data

> ***For every case:*** *(1) Is it real or a tracking artefact? (2) Decompose the metric tree. (3) 3–4 hypotheses, each with the one data cut that confirms or kills it. (4) Prioritise by size × ease. (5) Recommend + success metric + guardrail.*

1. Flipkart: conversion down 10% WoW

Hypothesis	Data that confirms or kills it
H1 Tracking or release bug	server vs client events, platform/app-version split
H2 Traffic mix shifted to low-intent users	conversion by channel, new vs existing
H3 Price or deal perception worsened	competitor price checks, discount depth, deal density
H4 Delivery promise or serviceability worsened	pincode-level serviceability, EDD
H5 Checkout or payment failures	step-wise funnel, payment success by gateway
H6 Seasonality or post-sale hangover	YoY, same window last year

Recommend: Check release/tracking first (cheapest), then channel mix; recommend fixing the largest confirmed leak. Success: conversion back to baseline, guardrail on AOV.

2. Hike Messenger: DAU down 12%, concentrated in 60–90 day users

Hypothesis	Data that confirms or kills it
H1 Network not formed	active contacts, groups per user: churned vs retained (find the elbow/magic number)
H2 Cohort quality (bad acquisition month)	retention by install month and channel
H3 Notification change or opt-outs	opt-out rate, push CTR, send volume before/after
H4 Cinema cannibalising or failing	DAU split by feature (messaging vs Cinema)
H5 Release regression	Android/iOS and app-version split, crash rate

Recommend: Validate the magic-number hypothesis first, then nudge below-threshold users with a holdout. Success: 60–90d retention lift; guardrail opt-outs and ratings.

3. Hike Cinema: watch time +20%, D30 retention flat

Hypothesis	Data that confirms or kills it
H1 Depth increase in already-retained users, no new habit	minutes per retained user, retained-viewer count
H2 Autoplay inflating passive watch	autoplay continuation, exit within 30s of next episode
H3 A few binge-heavy series driving it	watch-time concentration by series
H4 New-user hook weak	ep1 completion, ep2 within 48h by cohort

Recommend: Judge success on retained viewers and ep2-in-48h, not minutes. Cap autoplay passivity if H2 is true. Guardrail: uninstall and D30.

4. Groww: KYC-complete users aren't trading

Hypothesis	Data that confirms or kills it
H1 Funding friction	KYC→funding drop, bank-link failures
H2 Intent decay (long gap)	first-trade rate vs days since KYC
H3 Low-intent acquisition (referral/incentive)	first-trade rate by source
H4 Product complexity or fear	first-session event heatmap, support tickets
H5 Market sentiment	activation vs Nifty trend, watchlist adds

Recommend: Fix the biggest step leak (usually funding); test one change per variant plus a holdout. Success: KYC→first-trade within 7 days.

5. Groww: brokerage revenue down, new accounts up

Hypothesis	Data that confirms or kills it
H1 Regulation hit F&O frequency	orders per trader, segment mix around rule dates
H2 New accounts inactive	active traders ÷ accounts, by cohort
H3 Active traders left (losses, competitors)	churn by P&L bucket, account closures
H4 Mix shift to lower-yield products	revenue per order by segment
H5 Pricing or fee change	brokerage per order trend

Recommend: Separate regulation from cohort quality before reacting. Success: revenue per active trader and active ÷ accounts; do not push trading frequency for its own sake.

6. eBay: promo ran, GMV up, ROI unclear

Hypothesis	Data that confirms or kills it
H1 Subsidised existing intent (always-buyers)	holdout or geo-control incremental GMV vs redeemer GMV
H2 Pull-forward	post-promo dip, repeat rate vs control
H3 GMV inflated by shipping or tax	GMV ex-shipping, ASP
H4 Cannibalisation across categories	platform-level GMV, not just promo categories
H5 Discount dependency	baseline purchase frequency of promo cohort over time

Recommend: Re-state ROI on incremental GMV ex-shipping vs a holdout. Recommend targeting persuadables and capping discount depth. Guardrail: baseline frequency.

7. Plus / loyalty: enrollments up, engagement flat

Hypothesis	Data that confirms or kills it
H1 Incentive-led enrolment of low-intent users	activation by enrollment source
H2 Benefits not discovered	benefit page views, utilisation rate
H3 Benefits not valuable	perceived-value survey, burn vs earn rate
H4 Lapse offsetting enrollment	net member growth, tenure curve
H5 Benefit cost cannibalising margin	GMV per member vs benefit cost, vs non-member lookalikes

Recommend: Shift targets from enrollment to activation and benefit utilisation. Success: % active members and GMV per member vs lookalike non-members.

8. Expedia: search-to-book down, cancellations up

Hypothesis	Data that confirms or kills it
H1 Price volatility or inventory gaps	price change between search and checkout, availability rate
H2 Mix shift to longer lead times	conversion and cancel by lead-time bucket
H3 Hidden-fee or non-refundable UX	cancel reason codes, fee-reveal step drop-off
H4 Competitor or metasearch traffic	conversion by channel, price parity checks
H5 Seasonal pattern	YoY by destination

Recommend: Quantify which lead-time and fee-reveal buckets drive the drop; fix price consistency before UX. Success: search-to-book and net (post-cancel) bookings.

9. Quick commerce: orders per user down, AOV up

Hypothesis	Data that confirms or kills it
H1 Low-frequency users left (survivorship)	cohort frequency, active-base composition
H2 Delivery-fee threshold pushed bigger, rarer baskets	basket distribution around the threshold
H3 Availability or ETA worsened	fill rate, ETA, cancellations by dark store
H4 Competitor promo	share-of-wallet survey, app-switching signals

Recommend: Do not read AOV up as health; measure cohort frequency and fill rate. Success: orders per retained user, with availability as the guardrail.

10. Paywall launch (Hike Plus): revenue on target, rating dropped

Hypothesis	Data that confirms or kills it
H1 Survivorship: engaged users converted, free-tier metrics look worse	free-only cohort, pre vs post
H2 Paid users feel low value	reviews paid vs free, ad exposure to payers, feature usage
H3 Too many upsell prompts	prompt frequency vs opt-out, rating, sessions
H4 Performance regression	load time, crash rate, "slow" in reviews
H5 Free-user switching risk	uninstall and D30 by segment vs holdout

Recommend: Do not roll back on the headline drop; run the free-only cohort comparison, fix payer value and prompt frequency. Success: rating recovery plus payer retention.

11. Flipkart Plus: GMV per member falling

Hypothesis	Data that confirms or kills it
H1 Dilution: new members are lower-spend	GMV per member by enrollment cohort
H2 Active % dropping (lapse)	% active, tenure curve, streak breaks
H3 Fewer orders per active member	orders/active member by category, deal exposure
H4 AOV down (discount mix)	AOV, items per order, discount depth
H5 Benefit change or competitor	before/after benefit change, share-of-wallet signals

Recommend: Report GMV per member by cohort, not blended. Success: GMV per active member and activation by source; check benefit-cost margin.

12. Hike Messenger: invite-loop conversion dropping

Hypothesis	Data that confirms or kills it
H1 Invite prompt less visible after UI change	prompt impressions, CTR by app version
H2 Contacts-permission denial up	permission grant rate, new-install cohort
H3 Invitee onboarding friction	invite-click → install → signup funnel
H4 Spam perception / link-preview change	report rate, delivery/open rate of invites
H5 Incentive fatigue	invites per user, reward redemption trend

Recommend: Instrument the invite funnel step by step; roll back UI if H1 holds. Success: invites sent → installs per active user.

13. eBay: take rate up, active sellers down

Hypothesis	Data that confirms or kills it
H1 Fee increase pushed out casual sellers	active sellers by size segment, elasticity (DiD vs unchanged category)
H2 Mix shift to higher-fee categories	take rate by category, GMV mix
H3 Ads revenue inflating take rate	ads vs final-value-fee revenue split
H4 Sellers migrating to competitors	competitor registrations, cross-platform listing overlap
H5 Seasonal supply dip	YoY seller count

Recommend: Segment sellers by size and estimate elasticity before any further fee move. Success: active sellers by segment alongside revenue.

14. UPI: transaction failure rate spiking

Hypothesis	Data that confirms or kills it
H1 One bank or PSP degraded	failure rate by issuer/acquirer bank, response code
H2 App release regression	app-version and OS split, timeout rate
H3 Peak-load issue	failure by hour/day, TPS vs capacity
H4 Fraud-control tightening	declines by risk rule, false-positive review
H5 Network or device issue	geography, telco, device model

Recommend: Page the bank/PSP owner first if H1 holds; otherwise release rollback. Success: failure rate by issuer; guardrail on fraud losses.

15. Ads: eCPM down while impressions up

Hypothesis	Data that confirms or kills it
H1 Ad-load increase diluted quality	ad slots per session, viewability
H2 Fill from lower-priced demand	fill rate by demand source, bid density
H3 Advertiser budget shift	spend by advertiser/category, auction pressure
H4 Mix shift to low-value placements or geos	eCPM by placement, geography
H5 Seasonality	YoY, same-week last year

Recommend: Quantify ad load vs demand-side causes; roll back load if quality dilution is confirmed. Success: revenue per session, not eCPM alone.

5. Interview Q&A from prep chats

Hike: Senior Manager Analytics

Metrics and cadence

- **Framing:** define what "good" looks like before measuring it: north star → leading indicators → guardrails. Know vanity vs decision metrics.
- **Messenger:** north star DAU/MAU or Daily Sending Users; depth = messages per DAU; breadth = active groups; health = D7/D30.
- **Cinema:** weekly retained viewers; ep1 completion, ep2 within 48h (habit loop), episodes/session, cross-product overlap with Messenger DAU.
- **Guardrails:** support contact rate, notification opt-out, report/block rate, uninstall rate.
- **Cadence:** daily anomaly digest; weekly Mon analytics sync, Wed PM metric review (3 slides), Fri leadership digest; monthly cohort health (D1/D7/D30/D90); quarterly OKRs. One named PM owner per metric; written metrics glossary with SQL.

Retention logic for a messaging app

- Retention is social: find the **magic number** (elbow where retained vs churned diverge, e.g. active contacts ≥10), then nudge users toward it (start chats, invite friends, groups).
- Prove nudges with a randomised holdout of at-risk users (e.g. 20% vs 40% retention) and track downstream health (messages, opt-outs, ratings).
- **DAU drop checklist:** real or tracking? (server vs client events), YoY seasonality, release changelog, then segment: new vs existing, platform, geo, channel, cohort age, feature (messaging vs Cinema).
- **Activation funnel:** install → open → signup → permissions → contact import → first message. Benchmark first; test one change per variant plus a holdout; watch last-step guardrail (selection bias lowers it when more low-intent users enter).

Cinema, monetisation and experiments

- **Improve retention:** diagnose by curve position: D1 = onboarding/autoplay, D7 = series hook/notifications, D30 = content breadth/recommendations.
- **Watch time:** optimise *intentional* watch time; inputs are recommendation relevance, episode length fit, notification timing.
- **Monetise:** ads (eCPM, fill, retention guardrail), subscription (free-to-paid, paid D30), brand integrations. Pre-commit what "working" means before launch.
- **Paywall cannibalisation (3 types):** free→paid conversion (fine), switching to competitors (serious), engagement drop among payers. Compare the free-only cohort pre vs post to avoid survivorship bias.
- **Experiments gone wrong:** never redefine the north star after seeing results; do not extend a test because p was disappointing (peeking); a metric like D30 "app open" can be gamed by notification timing, so prefer "% sending ≥1 message in week 4"; network effects in messaging mean cluster randomisation.
- **90-day experimentation plan:** month 1 align on primary/secondary/guardrail + MDE; month 2 analytics sign-off via a tool such as Statsig; month 3 biweekly review of shipped / abandoned / no-ship decisions.

eBay: Manager Data Science, Pricing / Promotions / Incentives

Experimentation fundamentals

Concept	What to say
OEC	The single pre-committed metric that decides ship / no-ship (e.g. on-time delivery rate). One metric prevents cherry-picking.
Guardrails	Non-negotiable kill switches you must not break (seller defect rate, contact rate, label completion).
Unit of randomisation	Choose buyer, seller, session, geo or time by where interference occurs. Seller-side features → randomise sellers (less power, but clean).
Switchback	Alternate treatment by time window when supply is shared. Needs washout for carryover and enough switches.
Market-level holdout	Whole markets treated vs held out for platform-wide pricing. Match markets, use DiD; costly and few units.
Power analysis	Sample size ∝ variance ÷ effect². MDE is a business decision. GMV is high variance: use CUPED or longer runs. Run ≥1–2 full weeks.
Novelty / cannibalisation / seasonality	Check effect decay over the window; measure platform-level not feature-level; avoid peak season launches or hold a continuous holdout.

Pricing and take-rate problems

- **Measure a take-rate change:** seller-level A/B is off the table (sellers talk). Use category or geo rollout with DiD, or synthetic control; check parallel trends.
- **Estimate elasticity** (% change in listings ÷ % change in take rate): naive regression is endogenous. Triangulate DiD on historical rate changes, regression discontinuity at volume-tier thresholds, and IV (competitor fee shocks). Always segment: casual sellers are far more elastic than power sellers.
- **Evaluate a rate-card change:** resist one success metric: revenue, GMV, seller count, competitor signals; short vs long horizon; pre-agree rollback thresholds.
- **Seller promo design:** threshold promos invite bunching; measure additionality (share who genuinely changed behaviour) with RD or holdout. Promo ROI = (Incremental GMV × take rate) − discount cost.
- **Seller GMV down 8%:** GMV = Active sellers × Listings/seller × Conversion × ASP; find the moved factor, then check competitor migration.

Incentives analytics (eBay context)

- **Promo ROI layers:** gross (GMV, conversion) → net (incremental GMV × take rate − cost, CAC) → long term (retention, LTV, repeat rate).
- **Trap:** eBay GMV includes shipping and tax, so strip those before attributing lift. Discounting trains buyers to wait for sales (discount-dependency trap).
- **Response vs uplift model:** response asks "who will buy?"; uplift asks "who will buy *because of* the promo?". Four segments: persuadables (target), always-buyers and never-buyers (waste), sleeping dogs (avoid).
- **Evaluate uplift** with Qini curve / AUUC, not AUC. Training data needs a randomised past experiment.
- **Your churn-model story:** LightGBM propensity → random test/control split on at-risk users → measured reactivation, transactions, GMV, so the holdout isolated incremental lift. "In retrospect it was a response model; I would layer uplift within the at-risk group." Churn model = eligibility, uplift model = prioritisation.

Your impact project (STAR): Meta performance-marketing incrementality

- **S/T:** Flipkart spent heavily on Meta; last-touch attribution inflated ROI. Own a rigorous incrementality measurement.
- **A:** designed an 8% holdout, sized from minimum holdout needed for significance on customer conversion over 2- and 3-week windows; weekly readouts; stat-sig on incremental customers in ~2

weeks and incremental GMV in ~3; split by BU (each owns its spend), then by super-category for weak BUs.

- **R:** established true ROI, found Meta healthy overall → budgets scaled; flagged super-category targeting fixes for weak BUs. Add the ₹ spend and % scale-up to land the punchline.

Expedia: Senior Data Scientist, Loyalty Analytics

Churn propensity model

- Define churn operationally first (no booking in 90 days, points expiry, tier drop); the intervention sets the prediction horizon.
- Features: recency/activity, engagement depth (earn/burn, balance), trajectory (booking trend), product mix, lifecycle (tenure, tier anniversary). No post-prediction leakage.
- Baseline gradient-boosted trees + SHAP; handle imbalance; evaluate top-decile precision/recall and calibration (not AUC alone). Top-decile precision = churners among the 10% contacted; recall denominator = all churners.
- Deploy weekly scoring to a feature store; monitor score/feature drift and outcome feedback; retrain on drift or quarterly. Success = lift over the rule-based approach and ROI vs saved LTV.

Uplift: the T-learner

- Split data by treatment flag; train Model_T on treated, Model_C on control. Score any member through **both**; uplift = P(retain | treated) − P(retain | control). Can be negative (sleeping dogs).
- Example: Member A 0.82 vs 0.55 → +0.27 (target); Member B 0.91 vs 0.89 → +0.02 (skip despite high propensity).
- Weakness: two noisy estimates, high variance with small arms → X-learner, R-learner, causal forests. Requires randomised treatment history. Alternatives: S-learner, class transformation.

A/B test for a new tier benefit

- Clarify benefit, tier, goal. Metrics: primary (incremental bookings or tier retention), secondary (redemption, engagement), guardrails (revenue per booking, cancellations, support).
- Randomise at member level across brands, stratified by booking history; check household spillover (cluster if needed).
- Power with MDE agreed with PM/finance; run ≥2 booking cycles; keep a 5–10% permanent holdout for long-term LTV; analyse intent-to-treat; report absolute lift with CI, segments and revenue impact.

When you cannot randomise: which method?

Situation	Method	Key check
Hard threshold (e.g. 50 nights for Platinum)	Regression discontinuity	McCrary density test (no bunching), covariate smoothness
Treated and untreated groups, pre/post data	Difference-in-differences	Parallel pre-trends; member fixed effects
Single market / policy launch	Synthetic control	Pre-period fit, placebo tests, long pre-period
Observational, non-random selection	Propensity score matching	Covariate balance (SMD < 0.1), common support; cannot fix unobserved confounders — run sensitivity analysis

Gaps and bridging

If loyalty experience is thin, bridge: "The core problems map to retention modelling, reward optimisation and segmentation of engaged vs dormant customers; I have familiarised myself with earn/burn and tier elasticity." Prepare stories for: loyalty/retention model, experiment design, cross-functional influence, model in production, data governance.

6. One-page cheat sheet

Answer pattern

- **Clarify:** what decision does this inform? Real or tracking artefact? Benchmark and YoY first.
- **Decompose:** draw the metric tree; find the factor that moved.
- **Hypothesise:** 3–4 specific causes; for each, name the one data cut that confirms or kills it.
- **Prioritise:** size of impact × ease of checking.
- **Recommend:** action, owner, success metric, guardrail, and what the data will look like if we are wrong.

Leadership packaging (5 lines)

- What is happening (1 sentence).
- Why (1–2 sentences).
- What happens if we do not act (1 sentence).
- My recommendation (1 sentence).
- How we measure success (1 sentence).

Phrases that signal seniority

- "Before diving in, I'd check if this is data quality or a real signal."
- "I'd pre-commit the primary metric, MDE and kill threshold before launch."
- "The test did not fail, our measurement plan did; here is how we fix it."
- "A response model finds likely buyers; an uplift model finds persuadable ones."
- "Higher short-term metric, but it is borrowing engagement from future periods."
- "Selection bias at the top of the funnel can make the next step look worse even when the whole funnel improves."

Avoid

- Jumping to SQL or tools before framing.
- Redefining the north star after seeing results; extending a test to fish for significance.
- AUC as the only metric; accuracy for uplift models.
- Declaring success without a valid counterfactual.